

EN 513: Life Extension Engineering

Course Code	EN 513
Course Name	Life Extension Engineering
L-T-P-C	3-0-0-3
Intended for	UG/PG
Approval	9th Senate (May 2016)

Course Contents

- **Module I**
Inspection, monitoring and materials degradation, Structural integrity assessment (input parameters and assessment method) and life prediction, Risk assessment, life extension and economics. [10 Lectures]
- **Module II**
Life prediction for boiler components: Materials and damage mechanism, boiler tube failure mechanism, Header-damage mechanism, Damage mechanism in steam pipes, Damage and life assessment of boiler component, Extrapolation of statistics of part failures, Methods based on temperature estimation, Integrated methodology for life assessments.
Life assessment for steam-turbine components: Materials and damage mechanism, Service failure experience, Remaining life assessment: methods for rotors, Remaining life assessment of valves and steam chests, Remaining-life analysis of blades, Remaining-life assessment of bolts. [12 Lectures]
- **Module III**
Life assessment techniques for combustion-turbines: Life assessment techniques, Crack initiation assessment, Crack-propagation analysis for vanes, Crack tolerance of blades, Integrated methodology for life assessment. [8 Lectures]
- **Module IV**
Service life, reliability and maintenance of structures: Structural reliability and service life Time dependent reliability and service life. Probabilistic methods for service life estimation of engineering structures. [6 Lectures]
- **Module V**
Pipeline rehabilitation systems for service life extension: Extending service life, Trenchless renewal methods (TRMs), Selection of renewal method, Current design concepts for pipeline renewals, Emerging design concepts for pipeline renewal systems, Long-term testing. [6 Lectures]

Text and Reference Books

1. Viswanathan, Ramaswamy, **Damage mechanisms and life assessment of high temperature components**, ASM international, 1989.
2. **Proceedings of the Third International Conference on Engineering Structural Integrity Assessment** organised by The Dynamic Testing Agency in conjunction with AEA Technology, held at Churchill College, Cambridge, U.K., September 24-26, 1996 Editors: J.H. Edwards, P.E.J. Flewitt, B.C. Gasper, K.A. McLarty, P. Stanley and B. Tomkins, 1996.
3. Karbhari, Vistasp M., and Luke S. Lee, **Service life estimation and extension of civil engineering structures**, 1st edition, Woodhead Publishing Limited, Elsevier, 2010.